Amartha X GDG Jakarta Hackathon Project Proposal

Team Name, Project Title: Archive, KuBuku

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1. Problem Statement

Amartha's core mission is to **empower female micro-entrepreneurs (UMKM) in rural Indonesia**, an **unbanked** and **financially excluded** group, by providing them with access to **affordable microloans**. However, this mission faces a **barrier**: many of these women **lack formal credit histories**, and traditional credit scoring methods are **inadequate** for assessing their creditworthiness. This lack of structured **financial and behavioral data** from this ultra-micro segment makes it **difficult** for Amartha to build a reliable model for loan allocation.

Rural UMKM, often first-time digital users, face significant challenges in **managing** and **proving** the health of their cash flow. Existing accounting applications are overly complex, requiring manual inputs and high digital literacy, skills that many micro-entrepreneurs in rural areas lack. As a result, UMKM **struggle** to demonstrate the financial viability of their businesses, making it **difficult** to access loans for growth. Meanwhile, Amartha is **forced** to rely on subjective data from field agents, **losing** valuable insights that could be used to build alternative credit-scoring models and validate business income. This lack of reliable data also prevents Amartha from delivering timely, relevant financial literacy lessons, further deepening the knowledge gap.

This **deadlock** is more than a technical challenge, it is a **crisis of opportunity**. Amartha's ability to extend **meaningful financial support** to rural women entrepreneurs is **severely limited**. If this problem persists, the gap in financial inclusion will only continue to widen, leaving **millions of women** without the resources they need to thrive.

2. Proposed Solution/Concept

We propose **KuBuku**, an intelligent business assistant application specifically designed for Amartha's **"new digital user bases."** This is not just a bookkeeping app, KuBuku is a **data-generation tool** that transforms everyday language interactions whether in **formal Indonesian language**, **local dialects**, or **any language** in the world into **structured financial data**. Here are the key features of KuBuku:

1. **Quick Bookkeeping:** Users **simply speak** into the app (or type / take a photo of a receipt): AI will parse this into structured transaction entries.
2. **Personal Data Analyst:** Users can **ask in natural language**: "What sold the most this week?" or "How much profit did I make today?" The AI chatbot will provide simple answers, effectively delivering practical "financial literacy lessons."
3. **User Dashboard**: Users can **visually monitor** their business through daily/weekly/monthly sales graphs, automatic profit-loss summaries, best-selling product analysis, and inventory tracking. Data is displayed in real-time and easy to understand.
4. **Smart Opportunities:** AI proactively **analyzes user sales data** and provides relevant

recommendations for Amartha features (Working Capital, Savings (Celengan), Electricity Agent, AmarthaLink, Other Services):

5. **Amartha Validation Dashboard:** On the Amartha side, a dashboard will display structured bookkeeping data from all users. This provides **real-time data for "business income validation"** and becomes a very **powerful "alternative credit-scoring data."** This dashboard will also "flag" users who are "interested" in proactive offers, creating relevant recommendations for the Business Partner Team.

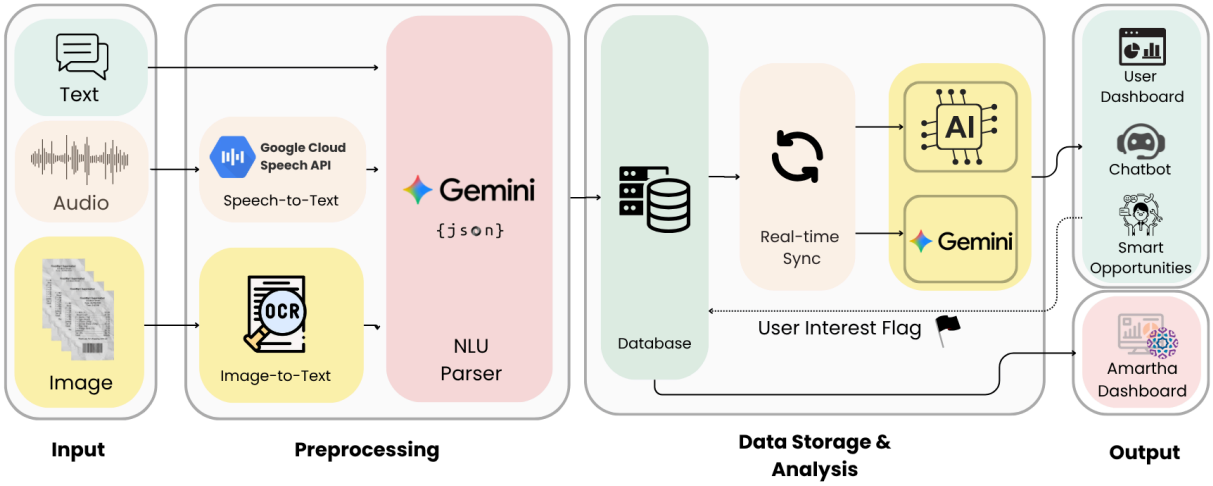

Image 1: KuBuku Technical Architecture

3. Target Users

Our solution has **two mutually beneficial** target users:

1. UMKM Actors: The core segment of Amartha: **female UMKM actors** in **rural areas,** who are often **"new digital user bases."**

- **Persona:** "Mrs. Siti, 45 years old, owner of a small grocery store and Amartha borrower. She is busy, finds apps complicated, and has no time to record sales. As a result, she struggles to prove her business cash flow."
- **Benefits:** KuBuku allows Mrs. Siti to record sales via text/voice/image in 5 seconds, helping her understand her business, and getting smart recommendations for other Amartha features that can advance her business.

2. Amartha: Amartha's Credit Analyst and Business Development Team.

- **Problems:** Currently, they face Amartha's core challenge, namely the **"lack of structured financial data"** from the ultra-micro segment and the urgent need for **"business income validation."**
- **Benefits:** Our app directly solves this problem. It transforms Mrs. Siti's voice input (unstructured data) into **vital structured financial data**. This gives Amartha **real-time data for "alternative credit-scoring models"** that are far more accurate

and reduces the risk of default.

4. High-Level Implementation Strategy

We will build a **prototype (MVP)** focused on the core workflow using **modern AI technology**.

Mobile Application (Frontend): Flutter for fast cross-platform development. The UI will be designed for maximum simplicity (large buttons, voice instructions).

Backend: Firebase (Firestore) for real-time database and fast authentication, or **Node.js/Python** (Flask) API hosted on **Google Cloud Platform.**

AI Architecture (Core):

1. **Voice Input: Google Speech-to-Text API** for accurate Indonesian language transcription.
2. **Logic & Analysis: Google Gemini API**. This is the core of our project and will be used for:
    - **Transaction NLU:** Parsing unstructured text into JSON
    - **Analyst Chatbot:** Answering user questions
    - **Generative AI:** Generating personalized, educational, and empathetic recommendation text for "Smart Opportunities" features, connected to Amartha products (Capital, Savings, AmarthaLink).
3. **Image Input: Google Vision AI** or **TensorFlow Lite** for OCR on receipts / bills.

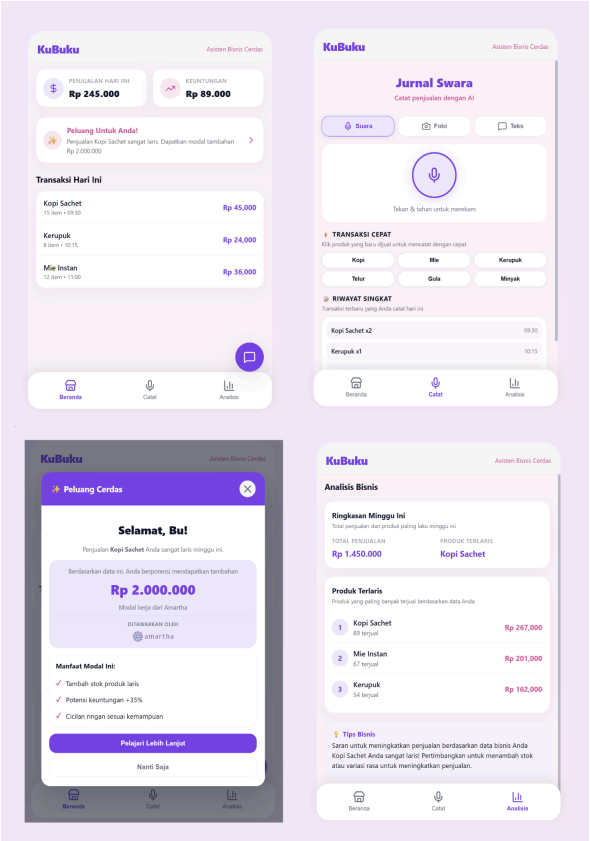

Image 2: KuBuku MVP Mock-up

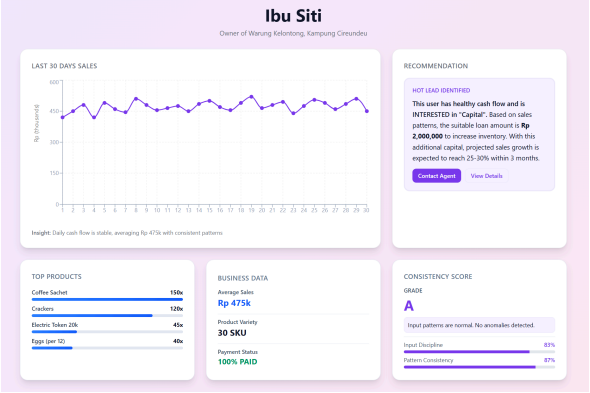

Image 3: Amartha Validation Dashboard UI

Amartha Dashboard (Web): React/Vue.js to visualize structured data from the backend.

MVP Scope: (1) Users can log in and (2) record transactions via text/voice/image processed by Gemini. (3) Users can view a simple daily sales dashboard. (4) Mockup of the "Proactive Nudge" flow. (5) Amartha's internal dashboard showing incoming raw data.